Fiscal Year:	FY 2009	**Task Last Updated:**	FY 11/10/2009
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Levitation Observation of Dendrite Evolution in Steel Ternary Alloy Rapid Solidification (LODESTARS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	04/03/2008	**End Date:**	04/02/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Szofran, Frank	**Contact Phone:**	256-544-7777
Contact Email:	frank.szofran@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hyers, Robert (University of Massachusetts, Amherst)		
Grant/Contract No.:	NNX08AL21G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Undercooling in containerless processing has shown promise in producing metastable phases and materials that exhibit improved chemical homogeneity and ultra-fine grain sizes. Stainless steel alloys require special handling in order to achieve a significant undercooling range. In particular, chromium is particularly susceptible to contamination through reaction with crucible material. When attempting to perform rapid solidification tests, the formation of active heterogeneous nucleation sites limits the undercooling that may be achieved. By using a containerless technique, this contamination is minimized. The focus of this program is to study of the effect of microgravity on metastable solidification. By observing changes in the mechanism for nucleation of the stable phase following primary recalescence from undercooled melts, the role of convection in phase selection may be evaluated. The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural		

	evolution. Experiments focus on the use of convection as a processing control parameter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Highlights of Work Accomplished During Grant Period 2008-09. • Joined ESA flight experiment groups (ThermoLab and MAGNEPHAS) with full membership as part of collaborative ISS MSL-EML. • Attended ThermoLab Program Workshop, KTH Stockholm Sweden 2 June 2008. • Attended ThermoLab Workshop with hardware developer, Friedrichshafen 14 April 2009. • Team meetings at DLR – 4 June 2008 and 6 July 2009 to discuss parameter set development and sample sharing philosophy. • Team meetings at TMS – with Fecht and Herlach February 2009. • The team was successful in proposing for inclusion of a LODESTARS steel ternary alloy sample as part of the Batch I sample list as part of the ThermoLab collaboration with Hans Fecht. We are responsible for producing this sample and leading the parameter set development effort for flight experiments. • The team was successful in being approved to share processing time on two MAGNEPHAS FeCo samples as part of the Batch I and Batch II sample list. These samples are not under our direct control and we will aid in developing processing parameter sets. • UMass hosted PI Matson as visiting professor during sabbatical period Jan-June 2009. • Matson organized 17th Symposium on Thermophysical Properties for Metallurgy in Boulder CO 21 June 2009 with Ivan Egry of DLR. • Matson and Hyers invited to participate as members of the ESA Solidification Topical Team. • Recruited grad student Xiao Ye (UMass), Paul Choufani (Tufts) and Matthew Sherman (Tufts) to conduct ground-based support experiments.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Egry I, Löhofer G, Matson D, Wunderlich R. "EML processing measurement techniques." High Temperature Material Processes, 2009, Accepted. , Nov-2009
Articles in Peer-reviewed Journals	Hyers RW, Rogers JR. "A review of electrostatic levitation for materials research." High Temperature Material Processes, 2009, Accepted. , Nov-2009
Papers from Meeting Proceedings	Hyers RW. "Anomalies in the thermophysical properties of undercool glass-forming alloys." 17th Symposium on Thermophysical Properties, Boulder CO, June 22, 2009. Proceedings. 17th Symposium on Thermophysical Properties, Boulder CO, June 22, 2009. , Jun-2009